

15.1 GAS TURBINE ENGINE Fundamentals.

Question Number. 1. On an axial flow, dual compressor forward fan engine, the fan turns the same speed as the.

Option A. low pressure turbine.

Option B. high pressure compressor.

Option C. forward turbine wheel.

Correct Answer is. low pressure turbine.

Explanation. Rolls Royce The Jet engine Page 6 refers.

Question Number. 2. A turbo jet engine gives.

Option A. large acceleration to a small mass of air.

Option B. large acceleration to a large weight of air.

Option C. small acceleration to a large mass of air.

Correct Answer is. large acceleration to a small mass of air.

Explanation. Rolls Royce The Jet engine Page 2/3 refer.

Question Number. 3. The basic gas turbine engine is divided into two main sections: the cold section and the hot section.

Option A. The cold section includes the engine inlet, compressor, and turbine sections.

Option B. The hot section includes the combustor, diffuser, and exhaust.

Option C. The hot section includes the combustor, turbine, and exhaust.

Correct Answer is. The hot section includes the combustor, turbine, and exhaust.

Explanation. NIL.

Question Number. 4. A jet engine derives its thrust by.

Option A. drawing air into the compressor.

Option B. impingement of the propelling gases on the outside air.

Option C. reaction of the propelling gases.

Correct Answer is. reaction of the propelling gases.

Explanation. Newtons third law applies- Rolls Royce The Jet engine Page 2/3 refers.

Question Number. 5. Which of the following might be used to identify turbine discharge pressure?.

Option A. Pt7.

Option B. Pt2.

Option C. Tt7.

Correct Answer is. Pt7.

Explanation. Jeppesen A&P Powerplant Textbook 3-5.

Question Number. 6. In a free turbine.

Option A. there is a clutch between compressor and power output shaft.

Option B. there is a direct drive with a free-wheel unit.

Option C. there is no mechanical connection with the compressor.

Correct Answer is. there is no mechanical connection with the compressor.

Explanation. Rolls Royce The Jet engine Page 5 refers.

Question Number. 7. Bernoulli's Theorem states that at any point in a flow of gas.

Option A. the static pressure and dynamic pressure are equal.

Option B. the static pressure is less than the dynamic pressure.

Option C. the total energy remains constant.

Correct Answer is. the total energy remains constant.

Explanation. See Chapter 2 of Rolls Royce The Jet Engine.

Question Number. 8. The working fluid of a gas turbine engine is.

Option A. gasoline.

Option B. kerosene.

Option C. air.

Correct Answer is. air.

Explanation. Kerosene provides the energy to drive the air.

Question Number. 9. Which statements are true regarding aircraft engine propulsion?

Option A. Turbojet and turbofan engines impart a relatively large amount of acceleration to a smaller mass of air.

Option B. In modern turboprop engines, nearly 50 percent of the exhaust gas energy is extracted by turbines to drive the propeller and compressor with the rest providing exhaust thrust.

Option C. An engine driven propeller imparts a relatively small amount of acceleration to a large mass of air.

Correct Answer is. An engine driven propeller imparts a relatively small amount of acceleration to a large mass of air.

Explanation. Jeppesen A&P Powerplant Textbook 3-43.

Question Number. 10. As subsonic air flows through a convergent nozzle the velocity.

Option A. decreases.

Option B. increases.

Option C. remains constant.

Correct Answer is. increases.

Explanation. Bernoulli's Theorem again.

Question Number. 11. In a twin spool compressor system, the first stage turbine drives the.

Option A. N2 compressor.

Option B. N1 and N2 compressors.

Option C. N1 compressor.

Correct Answer is. N2 compressor.

Explanation. Jeppesen A&P Powerplant Textbook 3-18.

Question Number. 12. At what point in an axial flow turbojet engine will the highest gas pressures occur?.

Option A. At the compressor outlet.

Option B. At the turbine entrance.

Option C. Within the burner section.

Correct Answer is. At the compressor outlet.

Explanation. Jeppesen A&P Powerplant Textbook 3-20.

Question Number. 13. Which of the following units are generally used to measure aircraft noise?.

Option A. Effective perceived noise decibels (E P N d B).

Option B. Decibels (dB).

Option C. Sound pressure.

Correct Answer is. Effective perceived noise decibels (E P N dB).

Explanation. Rolls Royce The jet engine page 199 refers.

Question Number. 14. The diffuser section is located between.

Option A. the burner section and the turbine section.

Option B. station No. 7 and station No. 8.

Option C. the compressor section and the burner section.

Correct Answer is. the compressor section and the burner section.

Explanation. Jeppesen A&P Powerplant Textbook 3-20.

Question Number. 15. If the LP shaft shears.

Option A. turbine runaway occurs.

Option B. compressor overspeed occurs.

Option C. compressor underspeed occurs.

Correct Answer is. turbine runaway occurs.

Explanation. The turbine drives the LP compressor or fan.

Question Number. 16. The term Pt7 means.

Option A. pressure and temperature at station No. 7.

Option B. the total pressure at station No. 7.

Option C. the total inlet pressure.

Correct Answer is. the total pressure at station No. 7.

Explanation. NIL.

Question Number. 17. What section provides proper mixing of the fuel and efficient burning of the gases?.

Option A. Diffuser section and combustion section.

Option B. Combustion section and compressor section.

Option C. Combustion section only.

Correct Answer is. Combustion section only.

Explanation. Jeppesen A&P Powerplant Textbook 3-21.

Question Number. 18. Of the following, which engine type would most likely have a noise suppression unit installed?.

Option A. Turboprop.

Option B. Turbojet.

Option C. Turboshift.

Correct Answer is. Turbojet.

Explanation. Jepperson Gas Turbine Powerplants Page 3-57 refers.

Question Number. 19. The pressure of supersonic air as it flows through a divergent nozzle.

Option A. decreases.

Option B. increases.

Option C. is inversely proportional to the temperature.

Correct Answer is. decreases.

Explanation. A&P Airframe Textbook Page 2-31 and 2-32.

Question Number. 20. The symbol for designating the speed of a LP compressor in a twin spool engine is.

Option A. N.

Option B. NG.

Option C. N1.

Correct Answer is. N1.

Explanation. Jepperson Gas Turbine Powerplants Page 12-13 refers to N1 for LP N2 for H P.

Question Number. 21. A turbojet engine is smoother running than a piston engine because.

Option A. the lubrication is better.

Option B. it runs at a lower temperature.

Option C. it has no reciprocating parts.

Correct Answer is. it has no reciprocating parts.

Explanation. NIL.

Question Number. 22. A gas turbine engine comprises which three main sections?.

Option A. Compressor, diffuser, and stator.

Option B. Turbine, compressor, and combustion.

Option C. Turbine, combustion, and stator.

Correct Answer is. Turbine, compressor, and combustion.

Explanation. NIL.

Question Number. 23. When a volume of air is compressed.

Option A. heat is gained.

Option B. no heat is lost or gained.

Option C. heat is lost.

Correct Answer is. no heat is lost or gained.

Explanation. Jepperson Gas Turbine Powerplants Page 2-18 refers - assuming adiabatic.

Question Number. 24. The pressure of subsonic air as it flows through a convergent nozzle.

Option A. increases.

Option B. remains constant.

Option C. decreases.

Correct Answer is. decreases.

Explanation. NIL.

Question Number. 25. If a volume of a mass of air is 546 cubic feet at 273K, at 274K it will be.

Option A. 2 cubic feet greater.

Option B. $\frac{1}{273}$ less by weight.

Option C. 2 cubic feet smaller.

Correct Answer is. 2 cubic feet greater.

Explanation. NIL.

Question Number. 26. In what section of a turbojet engine is the jet nozzle located?.

Option A. Exhaust.

Option B. Turbine.

Option C. Combustion.

Correct Answer is. Exhaust.

Explanation. NIL.

Question Number. 27. Newton's First Law of Motion, generally termed the Law of Inertia, states:.

Option A. To every action there is an equal and opposite reaction.

Option B. Force is proportional to the product of mass and acceleration.

Option C. Every body persists in its state of rest, or of motion in a straight line, unless acted upon by some outside force.

Correct Answer is. Every body persists in its state of rest, or of motion in a straight line, unless acted upon by some outside force.

Explanation. NIL.

Question Number. 28. A high bypass engine results in.

Option A. overall slower airflow and greater propulsive efficiency.

Option B. overall faster airflow.

Option C. greater propulsive efficiency.

Correct Answer is. overall slower airflow and greater propulsive efficiency.

Explanation. Jepperson Gas Turbine Powerplants Page 2-29 refers.

Question Number. 29. Bernoulli's Theorem states that at any point in a flow of gas.

Option A. the static pressure and dynamic pressure are equal.

Option B. the static pressure is less than the dynamic pressure.

Option C. the total energy remains constant.

Correct Answer is. the total energy remains constant.

Explanation. NIL.

Question Number. 30. The Brayton cycle is known as the constant.

Option A. temperature cycle.

Option B. mass cycle.

Option C. pressure cycle.

Correct Answer is. pressure cycle.

Explanation. NIL.

Question Number. 31. In a choked nozzle, velocity increases, and.

Option A. density decreases.

Option B. pressure decreases.

Option C. pressure increases.

Correct Answer is. pressure increases.

Explanation. Jepperson Gas Turbine Powerplant Page 2-23. As the nozzle goes sonic the pressure starts to increase as a result of the shock wave.

Question Number. 32. Using standard atmospheric conditions, the standard sea level temperature is.

Option A. 29°C.

Option B. 59°F.

Option C. 59°C.

Correct Answer is. 59°F.

Explanation. NIL.

Question Number. 33. Standard sea level pressure is.

Option A. 29.92 inches Hg.

Option B. 29.29 inches Hg.

Option C. 29.00 inches Hg.

Correct Answer is. 29.92 inches Hg.

Explanation. NIL.

Question Number. 34. The highest pressure in a gas turbine is.

Option A. at the nozzle exit.

Option B. at the burner exit.

Option C. just after the last compressor stage but before the burner.

Correct Answer is. just after the last compressor stage but before the burner.

Explanation. Rolls Royce The Jet Engine Page 15 refers.

Question Number. 35. The velocity of subsonic air as it flows through a convergent nozzle.

Option A. remains constant.

Option B. increases.

Option C. decreases.

Correct Answer is. increases.

Explanation. NIL.

Question Number. 36. A turboprop engine derives its thrust by.

Option A. impingement of the prop-wash on the outside air.

Option B. reaction of the prop-wash.

Option C. reaction of the propulsion gases.

Correct Answer is. reaction of the prop-wash.

Explanation. Newtons Third Law.

Question Number. 37. Adiabatic compression is.

Option A. an isothermal process.

Option B. one where there is an increase in kinetic energy.

Option C. one where there is no loss or gain of heat.

Correct Answer is. one where there is no loss or gain of heat.

Explanation. Sherwin and Horsley Thermodynamics Page 144 refers.

Question Number. 38. In a ducted fan engine, the fan is driven by the.

Option A. turbine.

Option B. air passing over the compressor.

Option C. accessory gearbox.

Correct Answer is. turbine.

Explanation. Jeppesen Aircraft Powerplant Page 2-9.

Question Number. 39. A modular constructed gas turbine engine means that.

Option A. all engines have a specific component layout.

Option B. the engine is constructed by the vertical assembly technique.

Option C. its major components can be removed and replaced without disturbing the rest of the engine.

Correct Answer is. its major components can be removed and replaced without disturbing the rest of the engine.

Explanation. The Dictionary of Aircraft Terms by Dale Crane has this definition.

Question Number. 40. The accessory gearbox of a high bypass engine is.

Option A. on the HP Compressor housing.

Option B. in the forward bearing housing.

Option C. attached to the turbine casing.

Correct Answer is. on the HP Compressor housing.

Explanation. Jeppesen Aircraft Powerplant Page 3-9.

Question Number. 41. On a gas turbine engine, what is the fan driven by?.

Option A. I P turbine.

Option B. LP turbine.

Option C. H P turbine.

Correct Answer is. LP turbine.

Explanation. Rolls Royce The Jet Engine Page 6 refers.

Question Number. 42. Which law relates to the kinetic, pressure, and potential energy in a fluid flow?.

Option A. Bernoulli's theorem.

Option B. Newton's laws.

Option C. Charles's law.

Correct Answer is. Bernoulli's theorem.

Explanation. The sum of the energies in a system is constant. so if one decreases another will increase.

Question Number. 43. The density of gas may be expressed as.

Option A. volume/weight.

Option B. weight/volume.

Option C. pressure/volume.

Correct Answer is. weight/volume.

Explanation. Basic Physics this one.

Question Number. 44. E S HP is.

Option A. Horsepower/efficiency.

Option B. Shaft horse power + exhaust efflux.

Option C. Power available at the turbine less the power required to drive the.

Correct Answer is. Shaft horse power + exhaust efflux.

Explanation. Effective Shaft Horse Power is the measure of total power of a turbo prop engine.

Question Number. 45. A divergent duct will cause subsonic flow to decrease in.

Option A. velocity, increase pressure.

Option B. velocity, pressure remains constant.

Option C. pressure, increase velocity.

Correct Answer is. velocity, increase pressure.

Explanation. Rolls Royce The Jet Engine page 13 fig 2-3 refers.

Question Number. 46. The Brayton cycle is.

Option A. the name given to the intermittent cycling of an electrical de-icing system.

Option B. the continuous combustion cycle taking place in a gas turbine engine.

Option C. the constant velocity cycle taking place in a gas turbine engine.

Correct Answer is. the continuous combustion cycle taking place in a gas turbine engine.

Explanation. The Brayton Cycle is also known as the constant pressure cycle.

Question Number. 47. The purpose of a diffuser is to.

Option A. increase the kinetic energy of the air.

Option B. induce a swirl to the air prior to combustion.

Option C. increase the static pressure of the air.

Correct Answer is. increase the static pressure of the air.

Explanation. Diffusers are always static divergent ducts.

Question Number. 48. On a triple spool engine, the first stage of turbines drive.

Option A. the LP compressor.

Option B. the HP compressor.

Option C. the I P compressor.

Correct Answer is. the HP compressor.

Explanation. Rolls Royce The Jet Engine Fig 2-5-2 refers.

Question Number. 49. Ram effect is.

Option A. the increase of dynamic pressure at the face of the compressor.

Option B. conversion of static pressure to kinetic pressure at the face of the compressor.

Option C. conversion of kinetic energy to pressure energy at the face of the compressor.

Correct Answer is. conversion of kinetic energy to pressure energy at the face of the compressor.

Explanation. The greater the ram effect the greater the efficiency of the propulsion system.

Question Number. 50. Which of the following statements is true on a high bypass ratio turbofan?.

Option A. Both the compressor and combustion system are larger than their turbojet equivalent.

Option B. The compressor assembly is larger and combustion chamber smaller than their turbojet equivalent.

Option C. Both the compressor and combustion chamber are smaller than the turbojet equivalent.

Correct Answer is. Both the compressor and combustion chamber are smaller than their turbojet equivalent.

Explanation. Smaller compressors and combustion chambers can be used on high bypass fans as they are more efficient than turbo jets.

Question Number. 51. In the dual axial flow or twin spool compressor system with a free power turbine, N_f would be an indication of.

Option A. turbine thrust indication.

Option B. first stage compressor speed.

Option C. free power turbine speed.

Correct Answer is. free power turbine speed.

Explanation. The free turbine drives the prop-shaft only.

Question Number. 52. A waisted drive shaft is primarily to.

Option A. achieve dynamic balance.

Option B. reduce weight.

Option C. provide a fuse if the driven component is overloaded.

Correct Answer is. provide a fuse if the driven component is overloaded.

Explanation. An example of this type of drive shaft is fitted between a gearbox and an I D G .

Question Number. 53. The 'core engine' or 'gas generator' is made up of the following components:.

Option A. Inlet, compressor, combustion chamber, turbine, exhaust.

Option B. Turbine, combustion chamber, compressor.

Option C. Compressor, turbine, exhaust, propelling nozzle.

Correct Answer is. Turbine, combustion chamber, compressor.

Explanation. The core engine is that which the primary airflow passes (the air that passes through the combustion chamber). The inlet is not included as the air is normally split down the bypass duct at the entrance to the compressor system.

Question Number. 54. The principle of jet propulsion is.

Option A. the calorific value of fuel burnt is equal to aircraft.

Option B. the interaction of fluids and gases.

Option C. every action has a equal and opposite reaction.

Correct Answer is. every action has a equal and opposite reaction.

Explanation. This is Newtons Third Law.

Question Number. 55. Boyle's law states that, at constant temperature, if a gas is compressed.

Option A. its absolute pressure is proportional to its volume.

Option B. its absolute temperature is proportional to it's volume.

Option C. its absolute pressure is inversely proportional to its volume.

Correct Answer is. its absolute pressure is inversely proportional to its volume.

Explanation. If the volume is reduced the pressure goes up.

Question Number. 56. What part of a jet engine has the most potential energy?.

Option A. Immediately after the combustion chamber.

Option B. Just before the combustion chamber.

Option C. Immediately after the HP compressor.

Correct Answer is. Just before the combustion chamber.

Explanation. Where the fuel (unburned) is pumped in.

15.2 Engine Performance.

Question Number. 1. 'Ram effect' due to aircraft forward speed will cause the efficiency of the engine to.

Option A. remain constant.

Option B. decrease.

Option C. increase.

Correct Answer is. increase.

Explanation. Ram effect improves compression ratio which improves thrust without using extra fuel
Rolls Royce The Jet engine Page 219 refers.

Question Number. 2. The efficiency of a gas turbine engine at altitude.

Option A. decreases.

Option B. remains constant.

Option C. increases.

Correct Answer is. increases.

Explanation. The compressor performs better at lower air temperatures(see Jeppesen Gas Turbines Page 2-33) thus improving thermal efficiency. Also lower air temperatures up to the tropopause assist in maintaining Propulsive Efficiency(whilst accepting that decreasing density decreases mass flow). Also note that operators fly at the tropopause whenever possible even for short flights for the best SFC.

Question Number. 3. Which statement is true regarding jet engines?.

Option A. At the higher engine speeds, thrust increases rapidly with small increases in RPM.

Option B. At the lower engine speeds, thrust increases rapidly with small increases in RPM.

Option C. The thrust delivered per pound of air consumed is less at high altitude.

Correct Answer is. At the higher engine speeds, thrust increases rapidly with small increases in RPM.

Explanation. Jeppesen A&P Powerplant Textbook 3-43.

Question Number. 4. Some turboprop and turbojet engines are equipped with two spool or split compressors. When these engines are operated at high altitudes, the.

Option A. low pressure rotor will increase in speed as the compressor load decreases in the lower density air.

Option B. low pressure rotor will decrease in speed as the compressor load decreases in the lower density air.

Option C. throttle must be retarded to prevent overspeeding of the high pressure rotor due to the lower density air.

Correct Answer is. low pressure rotor will increase in speed as the compressor load decreases in the lower density air.

Explanation. Jeppesen A&P Powerplant Textbook 3-18.

Question Number. 5. 'Ram effect' due to aircraft forward speed will cause the thrust of the engine to.

Option A. remain constant.

Option B. decrease.

Option C. increase.

Correct Answer is. remain constant.

Explanation. Thrust is constant, but efficiency will increase.

Question Number. 6. With a fixed throttle, and with increased massairflow, what happens to EPR?.

Option A. EPR goes up.

Option B. EPR remains constant.

Option C. EPR goes down.

Correct Answer is. EPR goes down.

Explanation. With increasing mass airflow with throttles fixed Ram pressure (Pt2) increases but the P7 pressure remains the same. Jeppesen Aircraft Gas Turbines page 12-18 refers.

Question Number. 7. At what stage in a gas turbine engine are gas pressures the greatest?

Option A. Compressor outlet.

Option B. Turbine outlet.

Option C. Compressor inlet.

Correct Answer is. Compressor outlet.

Explanation. NIL.

Question Number. 8. Increasing ram effect with increased speed.

Option A. reduces thrust due to reduced compressor efficiency.

Option B. increases thrust due to increased maximum airflow.

Option C. reduces thrust due to reduced turbine temperature.

Correct Answer is. increases thrust due to increased maximum airflow.

Explanation. Jepperson Gas Turbine Powerplants Page 2-35 refers.

Question Number. 9. The highest heat to metal contact in a jet engine is the.

Option A. burner cans.

Option B. turbine inlet guide vanes.

Option C. turbine blades.

Correct Answer is. turbine inlet guide vanes.

Explanation. NIL.

Question Number. 10. Which compressor type gives the greatest advantages for both starting flexibility and improved high altitude performance?.

Option A. Single spool, axial flow.

Option B. Dual stage, centrifugal flow.

Option C. Split spool, axial flow.

Correct Answer is. Split spool, axial flow.

Explanation. NIL.

Question Number. 11. Which of the following is the ultimate limiting factor of turbine engine operation?.

Option A. Compressor inlet air temperature.

Option B. Burner can pressure.

Option C. Turbine inlet temperature.

Correct Answer is. Turbine inlet temperature.

Explanation. NIL.

Question Number. 12. At altitude, idling RPM is.

Option A. same as at sea level.

Option B. higher than at sea level.

Option C. lower than at sea level.

Correct Answer is. higher than at sea level.

Explanation. Due to decreased density there is less resistance to rotation.

Question Number. 13. Thrust.

Option A. increases with high temperature.

Option B. increases with low temperature.

Option C. decreases with low temperature.

Correct Answer is. increases with low temperature.

Explanation. Jeppesen Gas Turbine Powerplants Page 2-33 Refers.

Question Number. 14. Which of the following variables affect the inlet air density of a turbine engine?.

Option A. Altitude of the aircraft, Ambient temperature.

Option B. Compression ratio, Turbine inlet temperature, Altitude of the aircraft, Ambient temperature.

Option C. Speed of the aircraft, Compression ratio, Turbine inlet temperature, Altitude of the aircraft.

Correct Answer is. Altitude of the aircraft, Ambient temperature.

Explanation. NIL.

Question Number. 15. The propulsive efficiency is.

Option A. low, with a low mass flow acceleration.

Option B. high, with a low mass flow acceleration.

Option C. high, with a high mass flow acceleration.

Correct Answer is. high, with a low mass flow acceleration.

Explanation. Jeppesen Gas Turbine Powerplants Page 2-37 Refers - a large mass of air moved slowly!.

Question Number. 16. The RPM for maximum power would be.

Option A. lower on a colder day.

Option B. lower on a hotter day.

Option C. greater on a colder day.

Correct Answer is. lower on a colder day.

Explanation. Jeppesen Gas Turbine Powerplant Page 3-16 refers.

Question Number. 17. How does engine thrust vary with temperature?.

Option A. Increase in temperature gives greater thrust because of low friction in compressors.

Option B. Low temperatures give low thrust.

Option C. Low temperature gives greater mass flow and therefore greater thrust.

Correct Answer is. Low temperature gives greater mass flow and therefore greater thrust.

Explanation. Jeppesen Aircraft Gas Turbine Power plant Page 2-33 Refers.

Question Number. 18. A method of comparing engine efficiencies is by comparing.

Option A. fuel consumption.

Option B. thrust to weight ratio.

Option C. specific fuel consumption.

Correct Answer is. specific fuel consumption.

Explanation. Jeppesen Aircraft Gas Turbine Power plant Page 7-3 Refers.

Question Number. 19. With a fixed throttle in a climb.

Option A. RPM will increase.

Option B. RPM will remain constant.

Option C. RPM will decrease.

Correct Answer is. RPM will increase.

Explanation. Jeppesen Gas Turbine Powerplants Page 3-15 refers.

Question Number. 20. The point of maximum velocity in the engine is in the.

Option A. exhaust exit nozzle.

Option B. combustion chamber.

Option C. nozzle guide vanes.

Correct Answer is. exhaust exit nozzle.

Explanation. Rolls Royce The Jet Engine page 15 refers.

Question Number. 21. At constant RPM, the pressure ratio of the compressor and the temperature rise across the compressor.

Option A. increases with height.

Option B. remains constant irrespective of height.

Option C. decrease with height.

Correct Answer is. remains constant irrespective of height.

Explanation. Jeppesen Aircraft Gas Turbine Powerplant Page 3-20 refers.

Question Number. 22. With the aircraft stationary, propulsive efficiency.

Option A. depends on RPM.

Option B. is minimum.

Option C. is maximum.

Correct Answer is. is minimum.

Explanation. Jeppesen Gas Turbine Powerplants Page 2-29 refers.

Question Number. 23. The efficiency of conversion of kinetic energy into propulsive work is a measure of.

Option A. mechanical efficiency.

Option B. propulsive efficiency.

Option C. thermal efficiency.

Correct Answer is. propulsive efficiency.

Explanation. Jeppesen Gas Turbine Powerplants Page 2-29 refers.

Question Number. 24. What effect does high atmospheric humidity have on the operation of a jet engine?.

Option A. Decreases compressor and turbine RPM.

Option B. Decreases engine pressure ratio.

Option C. Has little or no effect.

Correct Answer is. Has little or no effect.

Explanation. Jeppesen A&P Technician Propulsion Textbook 7-70.

Question Number. 25. Power is adjusted in a gas turbine engine by.

Option A. increasing fuel flow.

Option B. increasing air and fuel flow.

Option C. increasing airflow to the combustion chamber.

Correct Answer is. increasing air and fuel flow.

Explanation. You cannot have extra fuel without extra air.

Question Number. 26. The engine rating plug.

Option A. is permanently connected to the E E C.

Option B. is connected to the EPR transmitter.

Option C. is permanently connected to the Engine casing.

Correct Answer is. is permanently connected to the Engine casing.

Explanation. Jeppesen Aircraft Powerplant Page 7-21.

Question Number. 27. Flat Rated thrust is defined as.

Option A. the thrust at the ambient temperature point above which thrust drops below 100%.

Option B. that power achieved at idle RPM.

Option C. that power achieved at maximum EGT.

Correct Answer is. the thrust at the ambient temperature point above which thrust drops below 100%.

Explanation. Jeppesen Aircraft Powerplant Page 7-34.

Question Number. 28. Thrust rating on an FADEC controlled engine can be changed by.

Option A. varying the ballast resistor in the EGT system.

Option B. changing the engine rating plug.

Option C. varying the EPR datum plug.

Correct Answer is. changing the engine rating plug.

Explanation. Jeppesen Aircraft Powerplant Page 7-21 refers.

Question Number. 29. Propeller torque is analogous to.

Option A. engine RPM.

Option B. shaft horsepower.

Option C. propeller RPM.

Correct Answer is. shaft horsepower.

Explanation. Propeller torque is equal and opposite to SHP under steady state conditions.

Question Number. 30. The total power in a turboprop engine is the.

Option A. SHP.

Option B. BHP.

Option C. E S HP .

Correct Answer is. E S HP .

Explanation. E S HP = shaft horse power plus residual gas exhaust thrust.

Question Number. 31. In a dive, with the throttles fixed, the EPR will.

Option A. not change.

Option B. increase.

Option C. decrease.

Correct Answer is. decrease.

Explanation. Jeppesen Aircraft Gas Turbines page 12-18 refers.

Question Number. 32. With an increase in forward speed, the engine thrust.

Option A. decreases slightly but recover due to ram effect.

Option B. increases.

Option C. decreases.

Correct Answer is. decreases slightly but recover due to ram effect.

Explanation. Jeppesen Aircraft Gas Turbine Powerplant Page 2-35 refers.

Question Number. 33. The main factor considered when designing an engine is.

Option A. maximum fuel consumption.

Option B. maximum turbine temperature.

Option C. maximum tip speed.

Correct Answer is. maximum turbine temperature.

Explanation. The turbine is the most highly stressed component in the engine.

Question Number. 34. To ensure an engine maintains self sustaining speed.

Option A. idle remains same for any density.

Option B. idle increases with density decrease.

Option C. idle increases with density increase.

Correct Answer is. idle increases with density decrease.

Explanation. Rolls Royce The Jet Engine page 103 para 15 refers.

Question Number. 35. A factor that limits EGT is the.

Option A. jet pipe.

Option B. compressors.

Option C. turbine.

Correct Answer is. turbine.

Explanation. Rolls Royce the Jet Engine Page 13 refers.

Question Number. 36. Thrust will.

Option A. increase at high temperatures.

Option B. decrease at low temperatures.

Option C. increase at low temperatures.

Correct Answer is. increase at low temperatures.

Explanation. Higher density gives higher mass flow hence higher thrust.

Question Number. 37. Across the turbines, there is.

Option A. a general temperature rise.

Option B. a general temperature drop.

Option C. an isometric expansion.

Correct Answer is. a general temperature drop.

Explanation. Rolls Royce the Jet Engine Page 15 shows a temperature decrease across the turbines due to energy extraction.

Question Number. 38. If the throttle position remains constant.

Option A. with increasing OAT, RPM and TGT will increase.

Option B. with decreasing OAT, RPM will increase.

Option C. with increasing OAT, TGT will increase.

Correct Answer is. with increasing OAT, RPM and TGT will increase.

Explanation. As OAT increases the air is thinner RPM increases but thrust decreases due to the thin air. Extra fuel is required to increase thrust therefore TGT increases.

Question Number. 39. If an aircraft climbs with a fixed throttle position.

Option A. thrust decreases and RPM increases.

Option B. thrust and RPM remain the same.

Option C. thrust increases and RPM remains the same.

Correct Answer is. thrust decreases and RPM increases.

Explanation. Thinner air causes thrust to decrease and the compressor to speed up.

Question Number. 40. Ram pressure recovery will generally take effect at aircraft speeds of.

Option A. mach 1.

Option B. mach 0.1 - 0.2.

Option C. only when the aircraft is stationary with engines running.

Correct Answer is. mach 0.1 - 0.2.

Explanation. Jeppesen Gas turbine Powerplant page 3-2 refers. With the aircraft stationary and engines running intake pressure is negative. As the aircraft begins its take off run the pressure recovers to above ambient (ram recovery).

Question Number. 41. As the air is passed through the turbine, due to the convergent shape formed between adjacent blades.

Option A. pressure decreases, velocity increases, temperature increases.

Option B. pressure increases, velocity increases, temperature constant.

Option C. pressure decreases, velocity increases, temperature decreases.

Correct Answer is. pressure decreases, velocity increases, temperature decreases.

Explanation. Rolls Royce The Jet Engine Fig 2-5-1 refers.

Question Number. 42. The hottest component in a gas turbine engine is.

Option A. the nozzle guide vanes.

Option B. the turbines.

Option C. the combustion chamber.

Correct Answer is. the combustion chamber.

Explanation. Combustors have to withstand flame temperatures of 2000 degrees C.

Question Number. 43. The basic equation for thrust is.

Option A. $\text{thrust} = \text{force} * \text{acceleration}$.

Option B. $\text{thrust} = \text{mass} * \text{velocity}$.

Option C. $\text{thrust} = \text{mass} * \text{acceleration}$.

Correct Answer is. $\text{thrust} = \text{mass} * \text{acceleration}$.

Explanation. Newtons second Law.

Question Number. 44. To maintain the selected RPM of a gas turbine at altitude.

Option A. the pilot will have to throttle back.

Option B. more fuel will automatically be added.

Option C. the fuel will automatically be reduced as the aircraft climbs.

Correct Answer is. the fuel will automatically be reduced as the aircraft climbs.

Explanation. In a hydro mechanical engine the P1 capsule will sense increasing altitude and trim off the fuel.

Question Number. 45. The term Pb means.

Option A. burner pressure measured at the diffuser case.

Option B. burner pressure measured at the NGV.

Option C. burner pressure measured at the combustion chamber.

Correct Answer is. burner pressure measured at the combustion chamber.

Explanation. Burner pressure is the static pressure in the combustor can, used in some systems to regulate fuel flow.

Question Number. 46. Which of the following is not an engine rating?.

Option A. Maximum Continuous.

Option B. Idle.

Option C. Maximum Take Off.

Correct Answer is. Idle.

Explanation. NIL.

Question Number. 47. At higher than standard day ambient temperatures, compressor speed will be.

Option A. lower than standard day speed.

Option B. no different.

Option C. higher than standard day speed.

Correct Answer is. higher than standard day speed.

Explanation. The air is thinner at higher temperatures, therefore the compressor has less load to work against and goes faster. RR Jet Engine Fig 2-18 refers. Note that due to the max allowable EGT the engine will reach a limiting 'corner-point thrust' and fuel will be trimmed off to prevent any over boost or over temperature

Question Number. 48. Ram Recovery' is a measure of.

Option A. intake efficiency.

Option B. net thrust.

Option C. forward air speed.

Correct Answer is. intake efficiency.

Explanation. Ram Recovery is the ability of an intake to convert kinetic energy into useful pressure energy.

Question Number. 49. most likely parameter limiting the height at which a jet engine powered aircraft can operate would be insufficient.

Option A. lift to support the aircraft weight.

Option B. mass airflow to maintain 15:1 air/fuel ratio.

Option C. oxygen to support combustion.

Correct Answer is. oxygen to support combustion.

Explanation. The engine will flame out with insufficient oxygen.

Question Number. 50. With increasing ram effect.

Option A. turbine temperatures decreases.

Option B. propulsive efficiency decreases.

Option C. propulsive efficiency increases.

Correct Answer is. propulsive efficiency increases.

Explanation. Increasing Ram Effect increases the overall system pressure ratio, hence increasing propulsive efficiency.

Question Number. 51. Full reverse power is approximately.

Option A. 95% of forward thrust.

Option B. 75% of forward thrust.

Option C. 50% of forward thrust.

Correct Answer is. 50% of forward thrust.

Explanation. Jeppesen Aircraft Gas Turbine Powerplants page 3-52 refers.

Question Number. 52. The efficiency of a gas turbine would be greatest at.

Option A. cold temperatures.

Option B. low pressure.

Option C. hot temperatures

Correct Answer is. cold temperatures

Explanation. Maximum thermal efficiency is achieved at the tropopause due to that being the coldest ambient temperature achievable.

Question Number. 53. In a gas turbine engine, turbine section.

Option A. temperature decreases along with pressure and velocity.

Option B. velocity decreases and pressure increases.

Option C. velocity increases and pressure decreases.

Correct Answer is. velocity increases and pressure decreases.

Explanation. Pressure and temperature always go down in the turbine as velocity goes up.

Question Number. 53. In a gas turbine engine, turbine section.

Option A. temperature decreases along with pressure and velocity.

Option B. velocity decreases and pressure increases.

Option C. velocity increases and pressure decreases.

Correct Answer is. velocity increases and pressure decreases.

Explanation. Pressure and temperature always go down in the turbine as velocity goes up.

Question Number. 54. As air density changes the RPM of a gas turbine engine will change. How is RPM kept at a constant speed?.

Option A. It is not.

Option B. Automatically by a simple engine device.

Option C. Manually by the pilot.

Correct Answer is. It is not.

Explanation. Whilst maximum RPM's are limited by various devices RPM is free to wander with changing density. eg As you climb higher minimum idle will increase.

Question Number. 55. With fixed throttle and increasing altitude, the fan on a high by-pass engine will.

Option A. decrease RPM.

Option B. increase RPM.

Option C. stay the same RPM.

Correct Answer is. increase RPM.

Explanation. NIL.

Question Number. 56. Through turbine rotor blades, the pressure.

Option A. increases, temperature increases, velocity decreases.

Option B. decreases, temperature and velocity decreases.

Option C. remains constant, temperature increases.

Correct Answer is. decreases, temperature and velocity decreases.

Explanation. RR The Jet Engine (4th edition) fig 2-5-1 page 15 (working cycle and airflow).

Question Number. 57. Temperature of the mass airflow through a Gas Turbine Engine.

Option A. increases from the inlet, through the compressor and the diffuser into the burner, and decreases through the turbine into the exhaust.

Option B. increases from the inlet, through the compressor and remains constant through the diffuser and increases at the burner, and decreases through the turbine into the exhaust.

Option C. remains constant at the inlet, increases from the compressor and the diffuser into the burner, and decreases through the turbine into the exhaust.

Correct Answer is. increases from the inlet, through the compressor and the diffuser into the burner, and decreases through.

Explanation. NIL.